

TOP 50 BOARD-PATTERN CASE STUDIES

Chemistry Special | Made by Pragya (Thunderstudy)

CASE 1 | Colligative Properties

A student prepares two solutions: A (1M Glucose) and B (1M NaCl). Colligative properties depend on the number of particles.

- Q1. Higher boiling point?
- Q2. Greater freezing point depression?
- Q3. Higher Van't Hoff factor (i)?
- Q4. Higher osmotic pressure?

A1: NaCl A2: NaCl A3: NaCl (i=2) A4: NaCl

CASE 2 | Electrolysis

Electrolysis of acidified water is a decomposition reaction where gases are liberated at the respective electrodes.

- Q1. Gas at cathode?
- Q2. Gas at anode?
- Q3. Which gas volume is double?
- Q4. Type of reaction?

A1: Hydrogen A2: Oxygen A3: Hydrogen A4: Redox

CASE 3 | SN1 Reaction

Substitution Nucleophilic Unimolecular (SN1) proceeds via the formation of a carbocation intermediate.

- Q1. Most stable carbocation?
- Q2. Ideal solvent type?
- Q3. Order of reaction?
- Q4. Intermediate species?

A1: Tertiary (3°) A2: Polar Protic A3: First Order A4: Carbocation

CASE 4 | Chemical Kinetics

A catalyst pathway provides an alternate route for a reaction to proceed by lowering the energy barrier.

- Q1. Energy changed by catalyst?
- Q2. Affect on Equilibrium constant?
- Q3. Affect on Rate constant?
- Q4. Is the catalyst consumed?

A1: Activation Energy A2: No change A3: Increases it A4: No

CASE 5 | Transition Metals

Colors of transition metal complexes arise from electronic transitions between split d-orbitals.

- Q1. Primary reason for color?
- Q2. Theory explaining this?
- Q3. Which transition occurs?
- Q4. Color in Zn^{2+} ion?

A1: d-d transition A2: Crystal Field Theory A3: Electronic A4: Colorless (d^{10})

Rapid-Fire Board Concepts (Cases 6-50)

CASES 6-10: Organic Basics

- 6. Why are alkyl amines strong bases?
- 7. Oxidation of secondary alcohols?
- 8. Fuel cell advantage?
- 9. Strong field ligand effect?
- 10. RBCs in hypertonic solution?

A6: +I Effect A7: Ketone A8: High efficiency/Eco-friendly A9: Electron pairing A10: Shrink (Exosmosis)

CASES 11-20: Physical & Inorganic

11. Association factor (i)?
12. Arrhenius Plot Slope?
13. Allyl halide reactivity?
14. Grignard + Water?
15. Tollen's Test used for?
16. Iodoform Test species?
17. Phenol acid character?
18. Chiral carbon group count?
19. Strong oxidizing agent potential?
20. Molar Cond. on dilution?

A11: < 1 A12: $-E_a/2.303R$ A13: Resonance A14: Alkane A15: Aldehydes A16: Methyl Ketone
A17: Resonance stabilized Phenoxide A18: Four different groups A19: High positive E° A20: Increases

CASES 21-35: Organic Reactions

21. S_N2 characteristic?
22. Zr and Hf similarity?
23. Protein primary definition?
24. DNA sugar type?
25. Catalyst poisoning?
26. Order determination?
27. Dehydration ease order?
28. Williamson Synthesis produce?
29. Sandmeyer starting material?
30. Kolbe's major product?
35. d-orbital splitting?

A21: Inversion A22: Lanthanoid Contraction A23: Amino acid sequence A24: Deoxyribose A25: Activity loss
A26: Rate constant units A27: $3^\circ > 2^\circ > 1^\circ$ A28: Ethers A29: Diazonium Salt A30: Salicylic Acid
A35: Crystal Field Splitting

CASES 36-50: Advanced Concepts

- 36. Paramagnetism cause?
- 37. Ionic mobility?
- 38. SN1 vs SN2 intermediate?
- 39. Electrophile?
- 40. Nucleophile?
- 41. Atomisation enthalpy?
- 42. Why variable oxidation states?
- 43. Buffer role?
- 44. Apollo mission fuel?
- 45. Addition polymer?
- 46. Coordination Number?
- 47. Multidentate stability?
- 48. Ideal solution law?
- 49. Non-ideal cause?
- 50. RDS definition?

A36: Unpaired e^- A37: Conductivity A38: Carbocation A39: e^- acceptor A40: e^- donor A41: High in d-block
A42: $(n-1)d$ and ns participation A43: Resist pH change A44: Hydrogen A45: Polythene/PVC
A46: Total donor atoms A47: Chelate effect A48: Raoult's Law A49: H-bonding changes A50: Slowest step



Revision Completed! All the best for your Boards! - Thunderstudy